Huikang Jiang

Email: hj2742@columbia.edu | https://huikangjiang06.github.io

Education

Columbia University September 2025 – May 2029
B.A. in Computer Science and Applied Mathematics

Beijing National Day School September 2021 – May 2025
High School

Experience

Research Assistant at ARiSE Lab, Columbia University September 2025 – Present
- Advised by Professor Baishakhi Ray and Professor Simin Chen.
- AI-driven Software Engineering. Trustworthy Coding Agents, Monitoring and Verification Systems, Agent Training.

Research Assistant at DitecT Lab, Columbia University January 2026 – May 2026
- Advised by Professor Sharon Di.
- LLM post-training. Enhancing LLM-human behavioral consistency for social science research.

Research Assistant at PILLAR Lab, Peking University May 2026 – Present
- Advised by Professor Liangming Pan.
- Mechanistic Interpretability. Representation Analysis and Engineering, Natural Language Autoencoders, and SAEs.

Research Engineer, KAV Labs September 2026 – Present
- A startup based at Columbia, enterprise-scale software engineering agents with policy inference and formal verification.

Research

SIGIL: Securing Agentic Code Generation with Intent Code Graph Learning
Huikang Jiang, Nihal Singh*, Hudson Reich, Simin Chen, Baishakhi Ray*
Under review at ICLR 2027 [PDF]
- Key Words: Agents for Software Engineering, Prompt Injection, Model Hijack, Graph Neural Networks
- Construct LLM-derived semantic graphs linking issue subtasks to code-chunk diffs and classify them (via GNNs or manually extracted features) to flag injected instructions that hijack SWE agents into producing vulnerable patches.

Does Steering Break Your Model? A Multi-Dimensional Evaluation Suite for LLM Steering Methods
Haotian Yang, Huikang Jiang*, Yucheng Wu*, Wen-Jie Jiang, Chenpeng Wang, Yibin Lou, Liangming Pan*
Under review at ICLR 2027 [arXiv]
- Key Words: Representation Engineering, Activation Steering, Side Effects
- Develop a comprehensive evaluation framework for representation engineering that compares methods across effectiveness, side effects, robustness, generalization, and data efficiency over a range of intervention methods.

Award

International Olympiad in Artificial Intelligence (IOAI) Bronze Medalist, World Rank 99
China's National Olympiad in Artificial Intelligence (NOAI) High Distinction, National Rank 5
China Computer Federation, National Driverless Car Contest Distinction, National Rank 8
American Mathematics Competition (AMC) 130.5 Points, Honor Roll, Global Top 5%
American Invitational Mathematics Examination (AIME) 12 points, Global Top 5%
International Mathematical Modeling Challenge (IMMC) Meritorious Award

Service

Member, Science Committee of National Olympiad in Artificial Intelligence (China)
Nov 2025 – October 2026
- Designed contest questions and sample exams.

Examiner, USA-North America AI Olympiad January 2026 – August 2026
- Designed and validated contest questions and sample exams.

Programming Skills

Java, Python, PyTorch